Lab Experiment 11

Mechanical and Architectural Design Software

Learning Outcomes

By the end of these activities, students will be able to:
1. **Model a mechanical part in Autodesk Fusion** by creating a 3D solid from provided dimensions and applying proper constraints and parametric features.
2. **Generate an engineering drawing** of the modeled part with accurate orthographic views, dimensions, and annotations according to basic drafting standards.
3. **Design a studio apartment layout in SketchUp**, incorporating a combined living room, kitchen, and bedroom area with an adjacent bathroom based on given floor-plan dimensions.
4. **Apply measurement and scaling tools** in both Fusion and SketchUp to ensure dimensional accuracy and clear representation of spatial relationships.
5. **Compare mechanical and architectural modeling workflows**, identifying similarities and key differences in interface, drawing conventions, and output requirements.

Lab Exercise

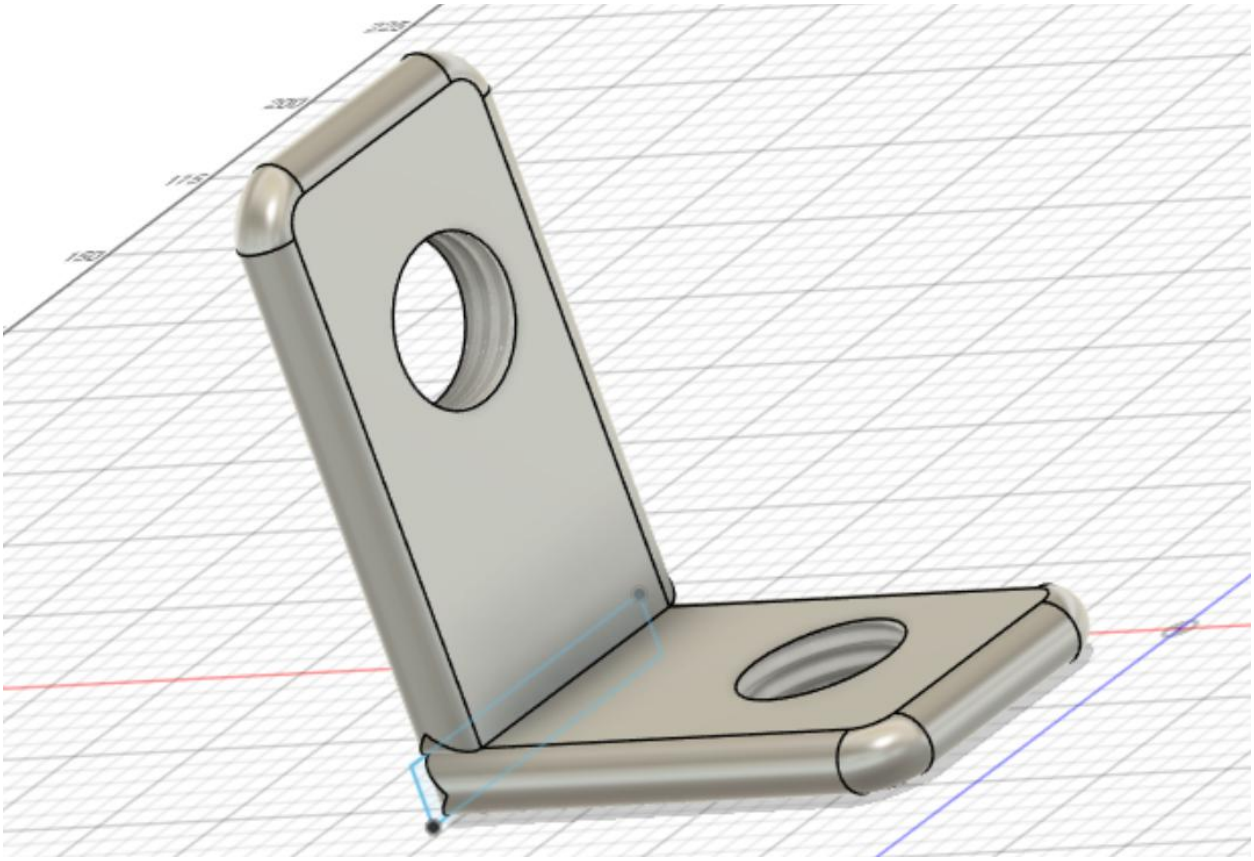
Objective:

To introduce students to mechanical and architectural design tools used in industry for 3D modeling and technical documentation. Students will model a simple mechanical part in Autodesk Fusion and create a fully dimensioned engineering drawing. They will then design a scaled studio apartment layout in SketchUp.

Instructions

Part A (11 points). Modeling in Fusion

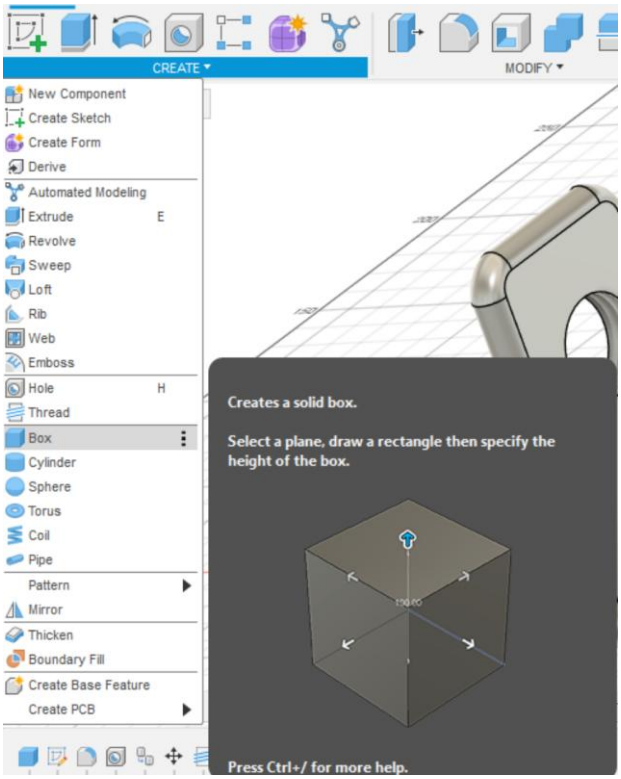
1. Model the following corner brace following the dimension drawing provided in pdf format (CornerBraceLab10)



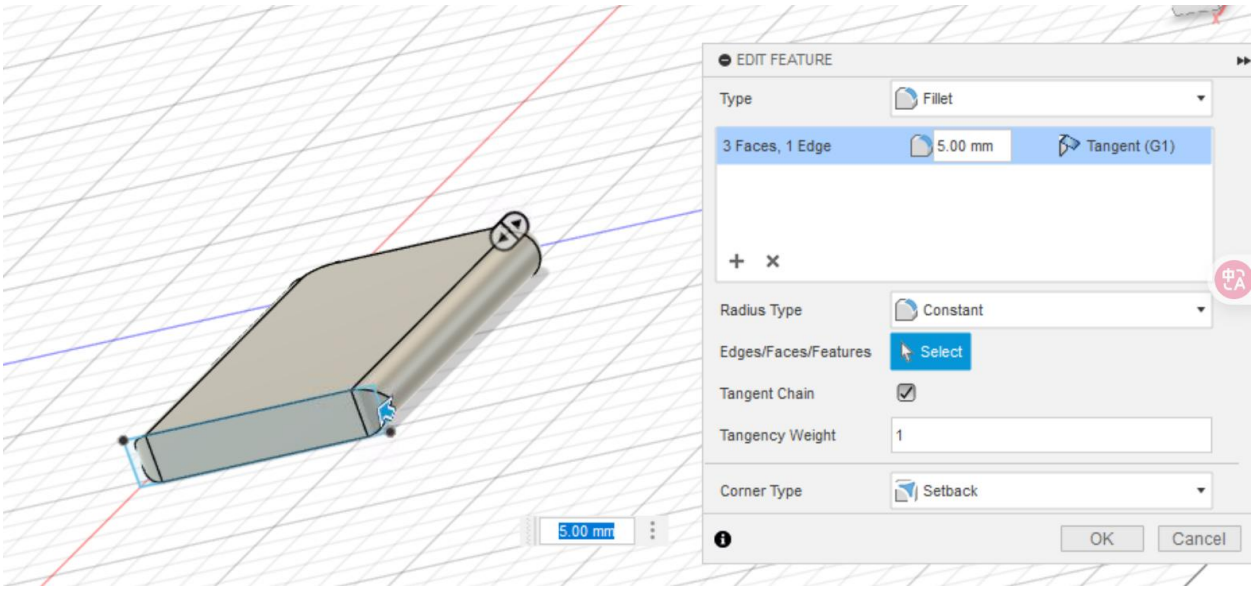
Hint:

This part consists of a rectangle with a threaded hole on each end, you can easily model the brace by completing these steps, make sure to reference the provided drawings for correct dimensioning during modeling

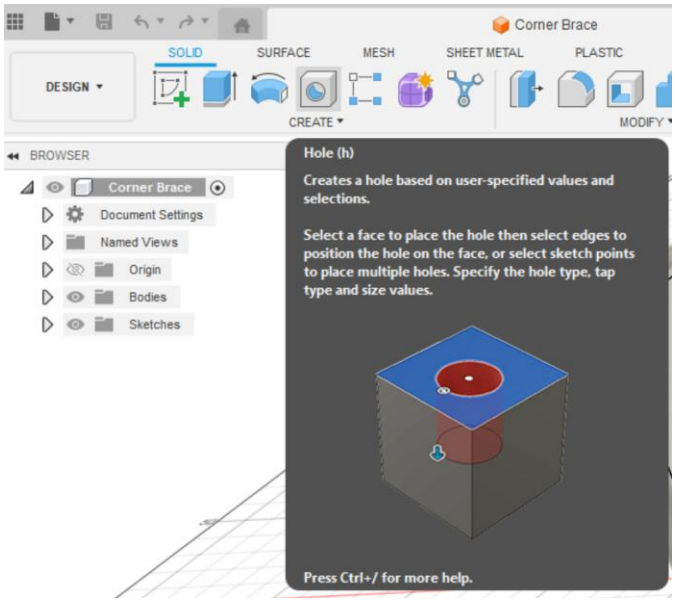
Step 1. Create a box normal to the surface



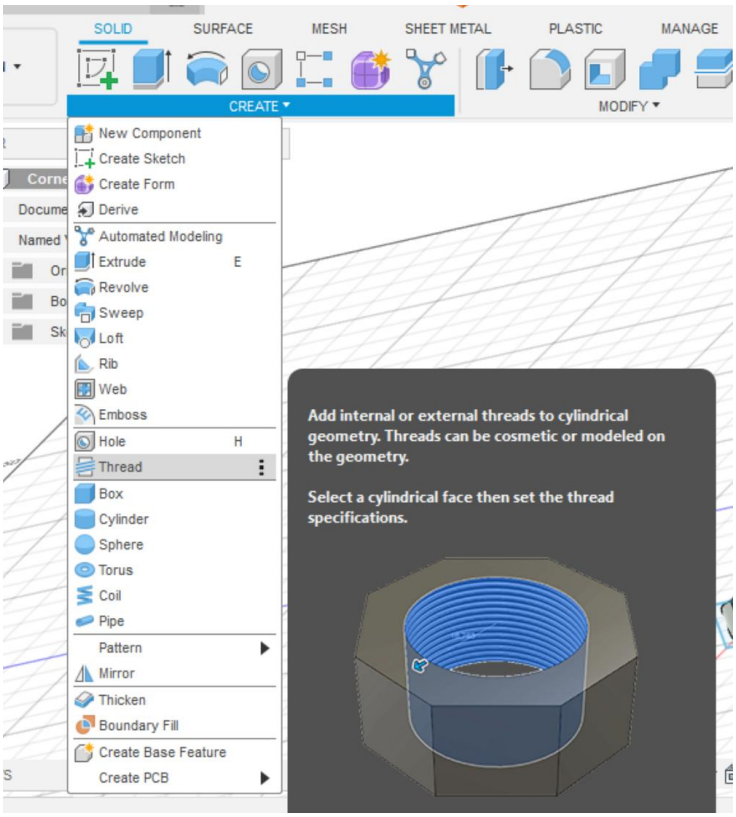
Step 2. Create a setback corner fillet on one of the box ends.



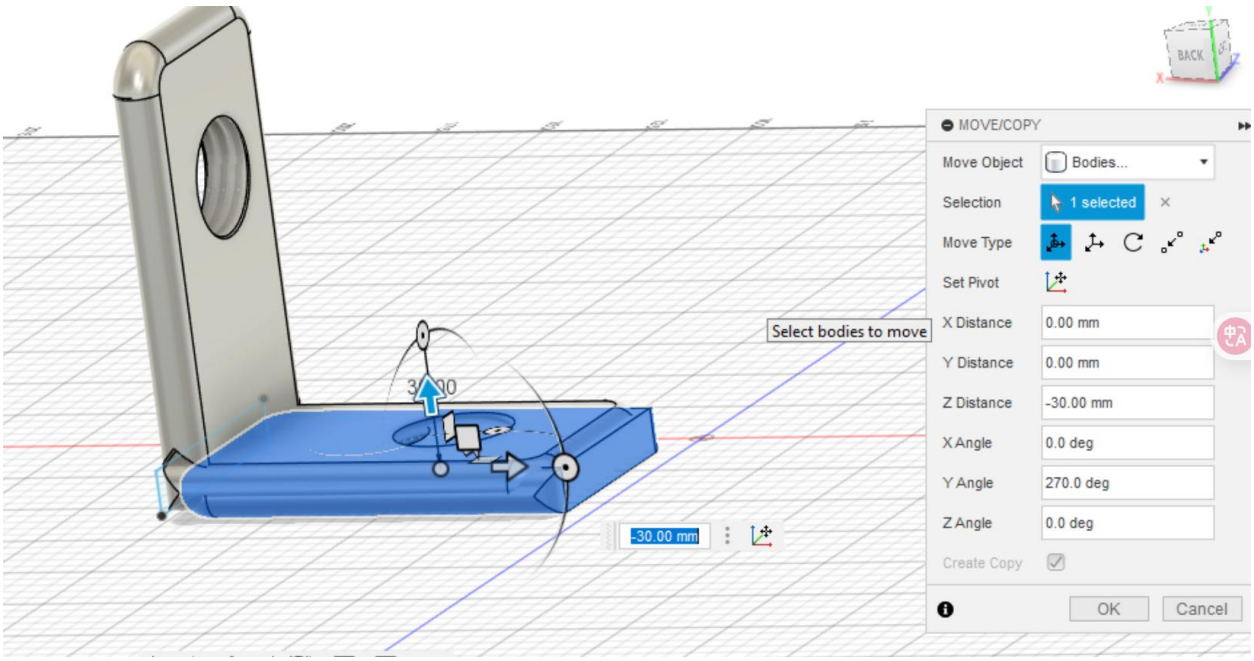
Step 3. Add a hole to the center of the box



Step 4. Add internal threads to the hole.



Step 5. Create a copy of the box and rotate it 270 degrees around the y-axis, you will see that the box was duplicated perpendicular to the surface box, making a L shaped frame, or corner brace.

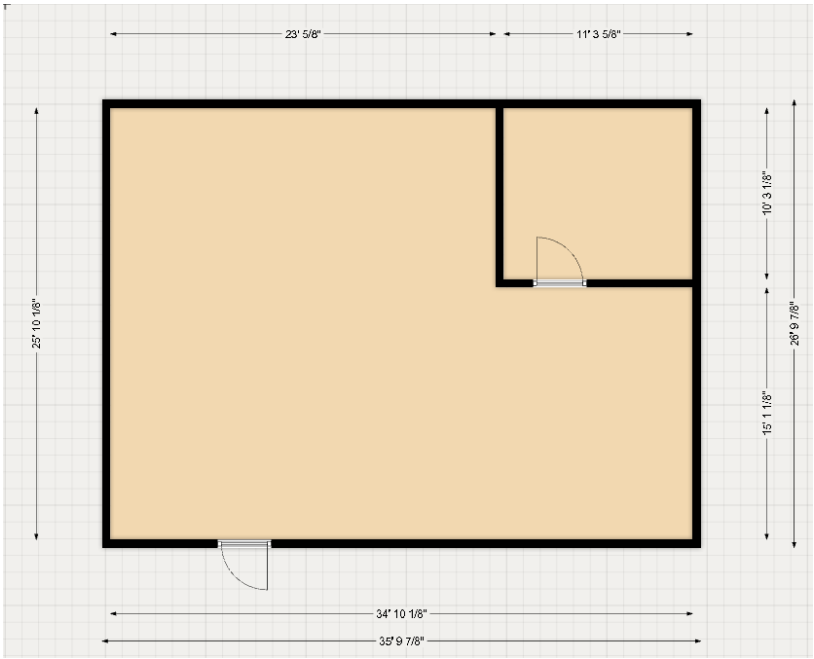


2. Once you modeled the part, create your own drawing sheet, **the drawing must have your name on the title block and display the appropriate dimensions in 4 different component views.**

Step 6. Export the drawing as a .pdf file

Part B (11 points). Modeling in Floorplanner

1. You are going to create a studio apartment referencing the following layout:



Dimensions don't have to be the same, this part of the lab is for you to familiarize yourself with the software.
Please note:

- Your layout must be like the one provided in the image, it must have a kitchen, living room and bedroom integrated into the same space, and a bathroom in the smaller room.
- Your final layout must also have objects in it (The sample layout is empty), like furniture and doors.

Deliverables:

Your final submission should include:

- A .**pdf** file of your **Fusion** 2D component drawing with appropriate dimensions.
- A .**png** file of your **Floorplanner** apartment layout with appropriate dimensions and modeled objects.

Grading Rubric

- **Part 1 -Fusion model**
  - Proper Title block formatting (0-4 points)

    - Proper dimensioning (0-4 points)
    - Model replicates the provided sample (0-4 points)

- **Part 2 - Floorplanner layout**
    - Layout replicates the provided sample (0-6 points)
    - Enough objects are present to differentiate the different zones (0-5 points)
- **Total points:** 20 + 2(extra)